18.Implement Perceptron based IRIS classification

CODING:

```
from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler

from sklearn.linear_model import Perceptron

from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

iris = load_iris()

X = iris.data

y = iris.target

print("Iris Dataset")

print("------------")

print("Number of samples:", X.shape[0])

print("Number of features:", X.shape[1])

print("Classes:", iris.target_names)

X_train, X_test, y_train, y_test = train_test_split(

    X, y,

    test_size=0.2,

    random_state=42,

    stratify=y

)

scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)

X_test = scaler.transform(X_test)

model = Perceptron(

    max_iter=1000,

    eta0=0.1,

    random_state=42
```

```
)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print("\nPerceptron Classification Results")
print("---------------------------------")
print("Actual values   :", y_test)
print("Predicted values:", y_pred)
print("Accuracy        :", accuracy)
print("Accuracy (%)    :", accuracy * 100)
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
print("\nClassification Report:")
print(classification_report(
    y_test,
    y_pred,
    target_names=iris.target_names
))
new_flower = [[5.1, 3.5, 1.4, 0.2]]
new_flower_scaled = scaler.transform(new_flower)
prediction = model.predict(new_flower_scaled)
print("\nNew Iris Flower")
print("----------------")
print("Sepal Length :", new_flower[0][0])
print("Sepal Width  :", new_flower[0][1])
print("Petal Length :", new_flower[0][2])
print("Petal Width  :", new_flower[0][3])
print("Predicted Class:", iris.target_names[prediction[0]])
```

OUTPUT:

```
Iris Dataset
------------
Number of samples: 150
Number of features: 4
Classes: ['setosa' 'versicolor' 'virginica']

Perceptron Classification Results
---------------------------------
Actual values   : [0 2 1 1 0 1 0 0 2 1 2 2 2 1 0 0 0 1 1 2 0 2 1 2 2 1 1 0 2 0]
Predicted values: [0 2 1 1 0 2 0 0 2 1 2 2 2 2 0 0 0 1 1 2 0 2 1 1 2 2 1 0 2 0]
Accuracy        : 0.8666666666666667
Accuracy (%)    : 86.66666666666667

Confusion Matrix:
[[10  0  0]
 [ 0  7  3]
 [ 0  1  9]]

Classification Report:
              precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        10
  versicolor       0.88      0.70      0.78        10
   virginica       0.75      0.90      0.82        10

    accuracy                           0.87        30
   macro avg       0.88      0.87      0.87        30
weighted avg       0.88      0.87      0.87        30


New Iris Flower
----------------
Sepal Length : 5.1
Sepal Width  : 3.5
Petal Length : 1.4
Petal Width  : 0.2
Predicted Class: setosa
```